

Amortized gravitational wave inference with Dingo

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Outline for this morning

- **Presentation (now):**
 - Normalizing flows
 - Diagnostics (PP plots, importance sampling)
 - Dingo
- **Tutorial 1:** Galactic binaries example (with narrow prior)
- **Tutorial 2a:** Covering the full prior
- **Tutorial 2b:** Using Dingo

Neural posterior estimation

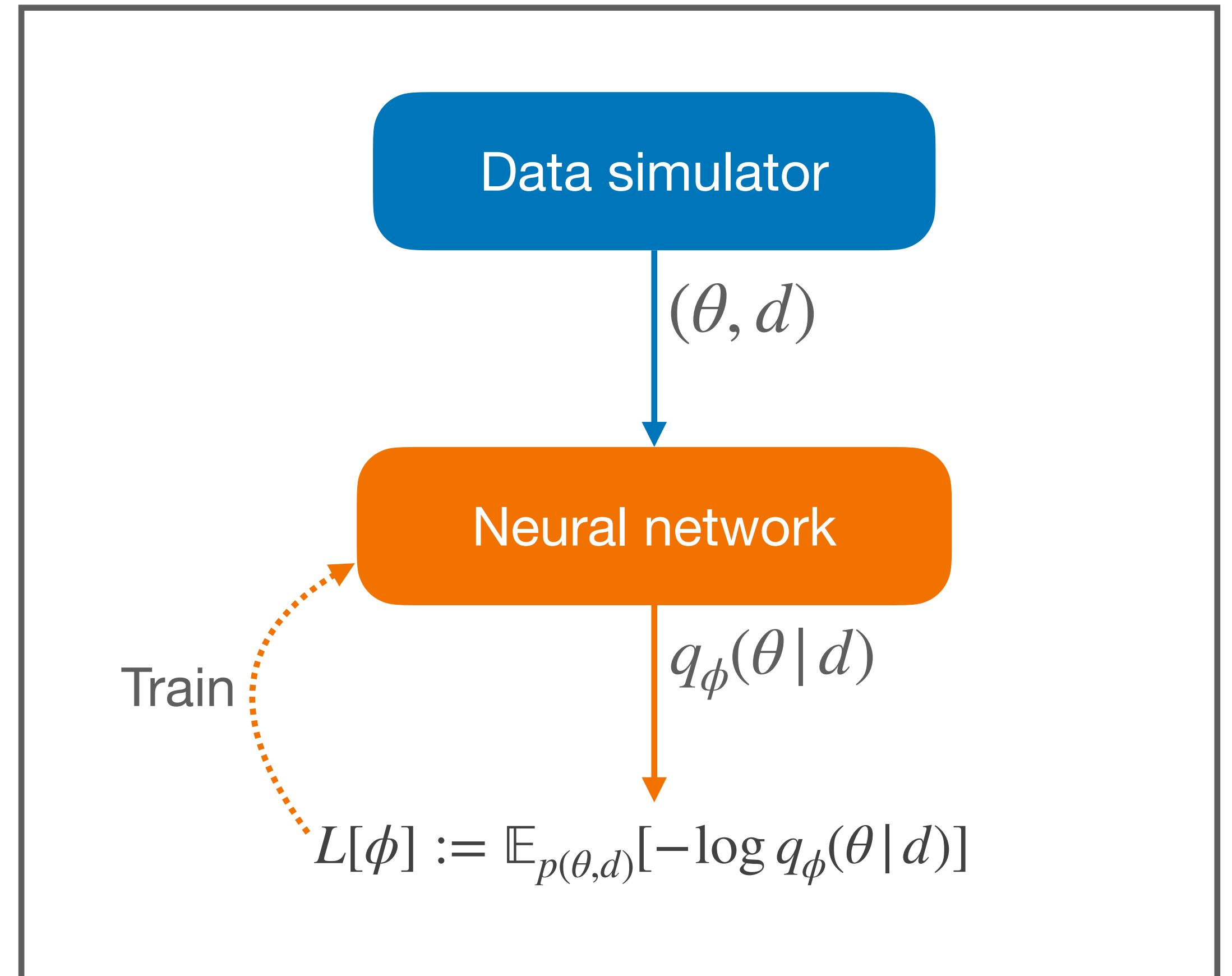
Recall from yesterday

- Replace likelihood evaluation with simulation:
Generate synthetic data $d \sim p(d | \theta)$.

- Use these simulations to train neural networks to infer parameters:
Learn the posterior,

$$q_\phi(\theta | d) \approx p(\theta | d)$$

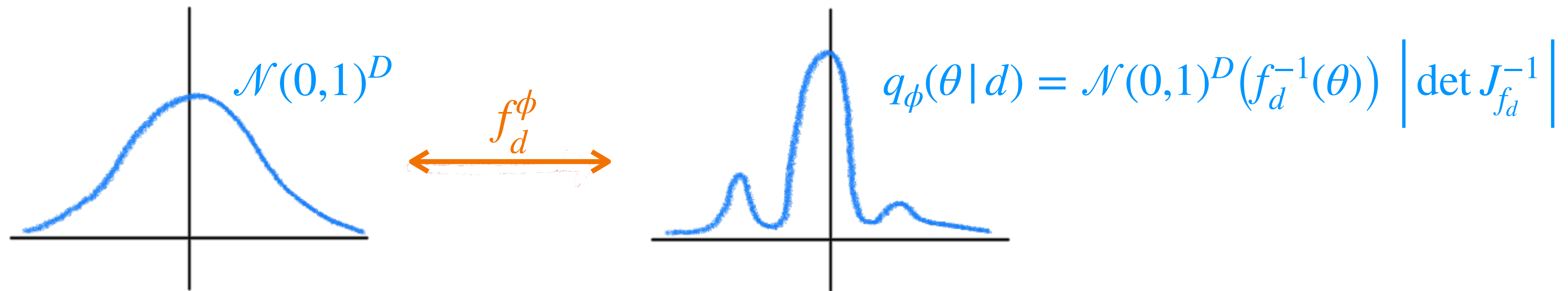
(No need to know the likelihood explicitly — works even when $p(d | \theta)$ intractable.)



Normalizing flows

Definition

- Represent a complex distribution q using an invertible mapping $f_d^\phi : u \mapsto \theta$ from a simpler distribution:

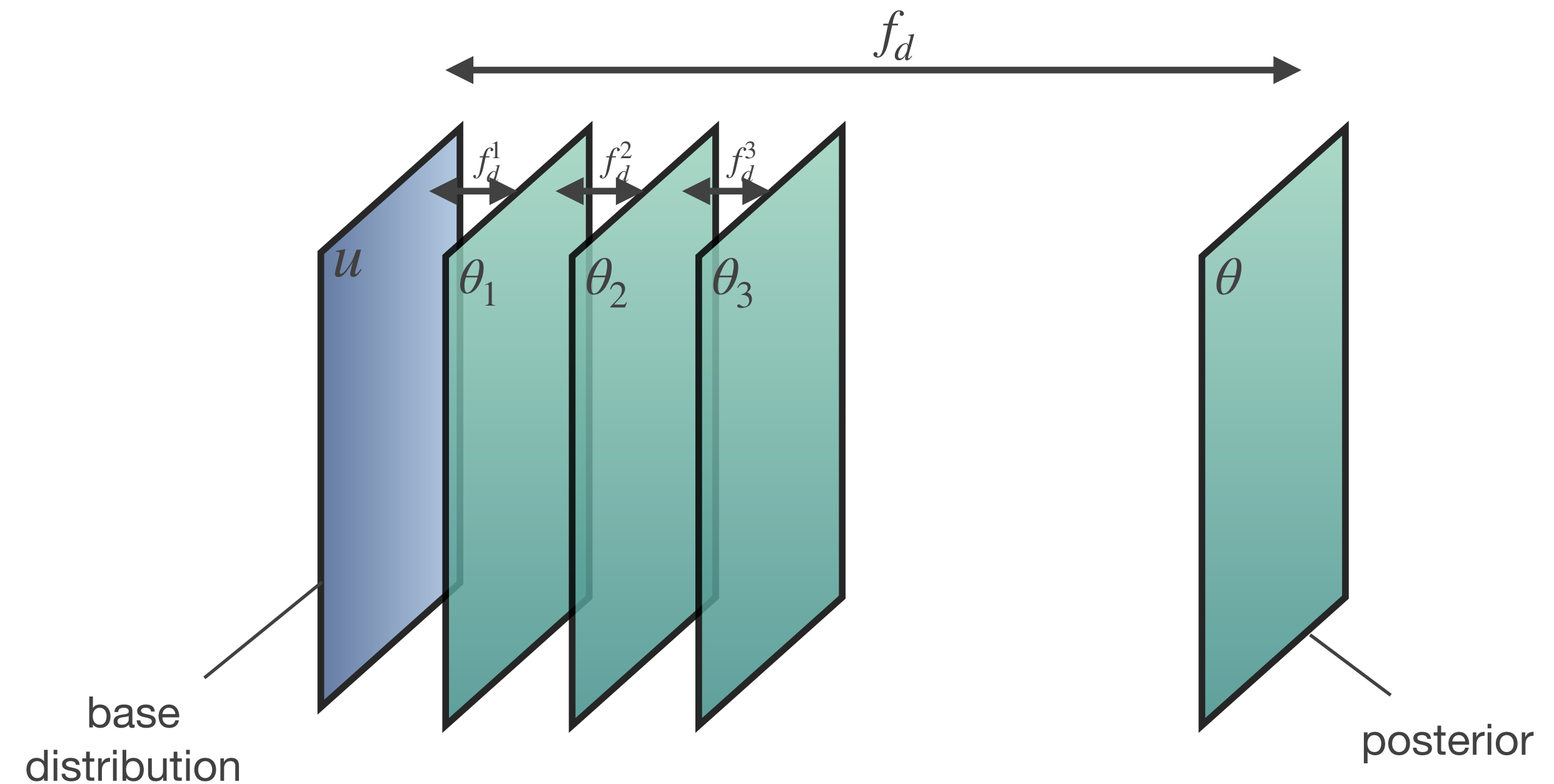


- f_d^ϕ parametrized using neural networks. Required to be invertible with simple Jacobian determinant. This allows for direct density evaluation, but at the cost of a constrained architecture.

Discrete normalizing flows

vs continuous flows

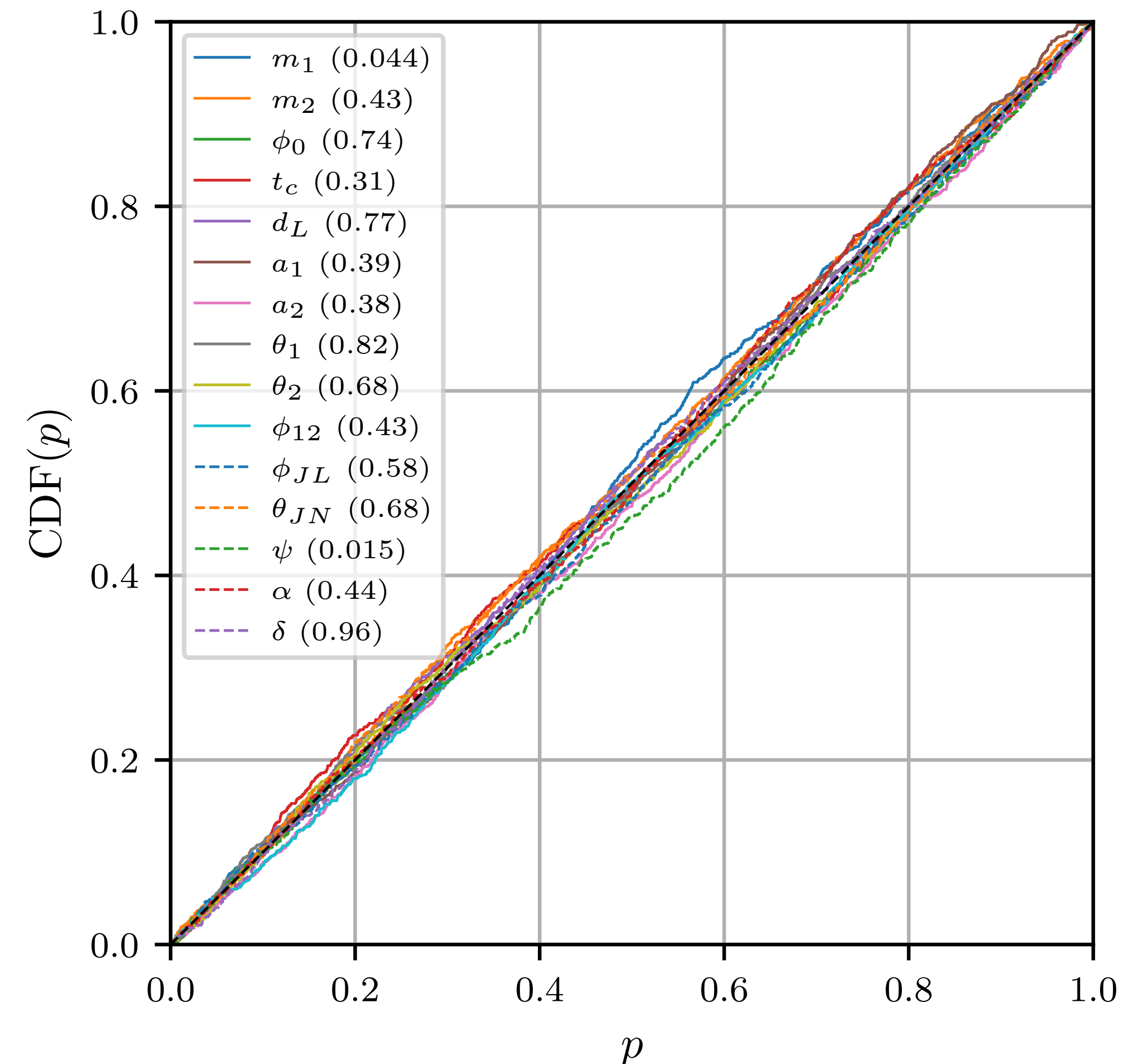
- Typically f_d^ϕ given as the composition of a sequence of parametrized transforms (affine maps, splines), which is why this is known as a **discrete flow**.
- The advantage is that both **sampling and density evaluation are fast**. Continuous flows require many evaluations to integrate ODE.
 - Better for time-critical applications like binary neutron star inference.
 - Allows for negative log likelihood training, easy importance sampling
 - Works well for moderate parameter dimension (~tens), and when this is known in advance.



Validating results

P—P plots

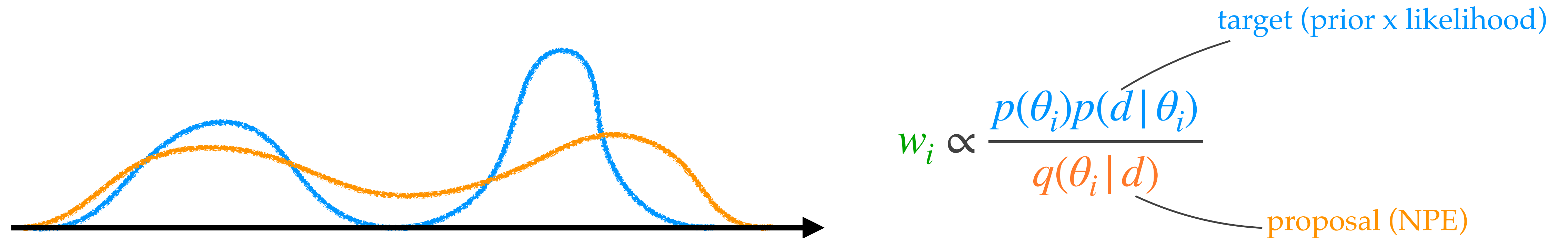
- **P-P plots** test whether truth values of simulated data sets are appropriately distributed within posteriors.
- Compute percentile score of each truth value within its 1D marginal. These should be uniformly distributed over $[0,1]$.
- Plot the CDF. Consistent with diagonal if posteriors are well-calibrated.
- Necessary (calibration on average over the prior), but not sufficient (says nothing about any single event)



Validation

Neural importance sampling

- For individual events, if a likelihood is available, can **importance sample** to the true posterior.

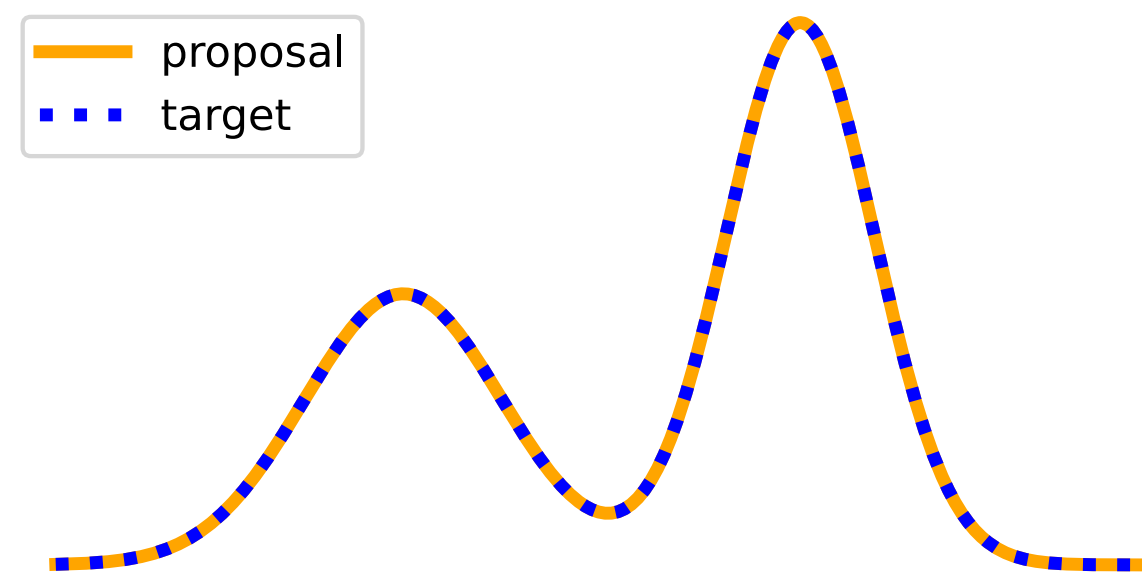


- Mean of the weights gives the **Bayesian evidence**: $p(d) = \frac{1}{n} \sum_{i=1}^n w_i$
- Variance gives the **sampling efficiency** $\epsilon = \left(\sum_i w_i \right)^2 / \left(n \sum_i w_i^2 \right)$

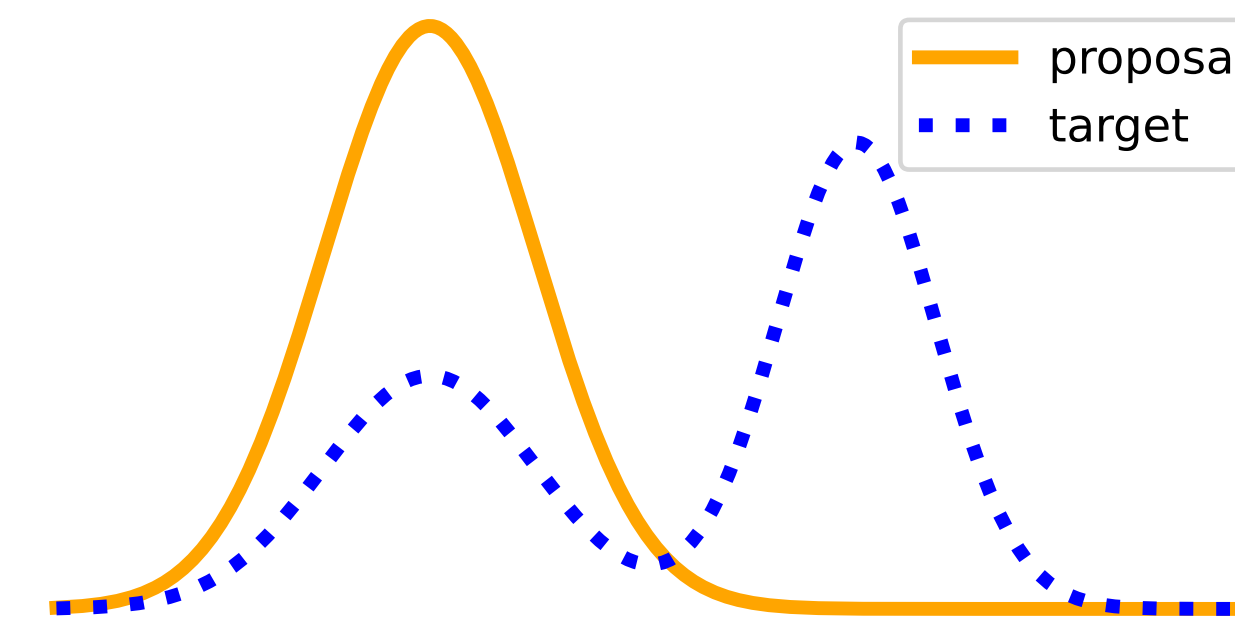
Neural importance sampling

- Performance depends on how good the proposal is

$$w_i = \text{constant} \implies \epsilon = 1$$



$$\text{large variance} \implies \epsilon \approx 0$$

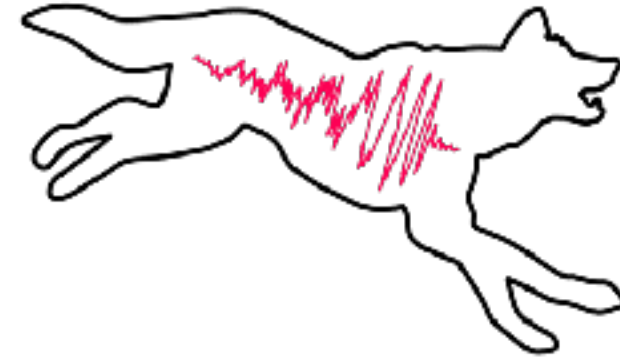


- NPE proposals have **mass-covering property**, making them well-suited to IS

$$L[\phi] \sim D_{\text{KL}}(p \mid q) = \int d\theta p(\theta \mid d) \log \frac{p(\theta \mid d)}{q_\phi(\theta \mid d)} \longleftarrow \text{Penalty in loss if } q_\phi \text{ misses a mode}$$

- Surprisingly, flow matching also gives mass covering, even without NLL loss.

DINGO



Code: <https://github.com/dingo-gw>

Docs: <https://dingo-gw.readthedocs.io/en/latest/>

What does it do?

1. Generate training datasets

- a. Intrinsic waveforms (polarizations), specified by masses, spins, inclination, and phase.
- b. Noise power spectral densities, with a distribution often corresponding to an observing run.

2. Train the network

- a. Construct network based on specified configuration.
- b. During training, sample extrinsic parameters (sky position, distance, time of coalescence) and noise realizations, and construct simulated data.

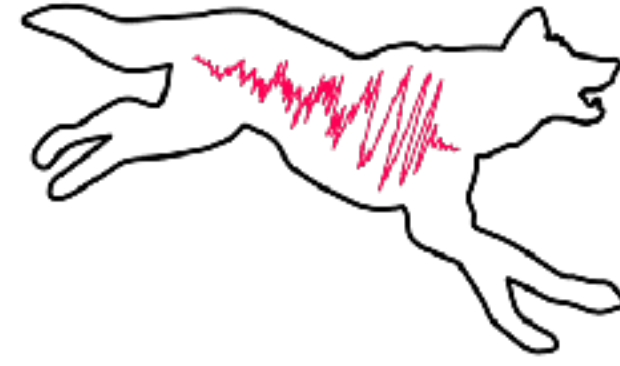
3. Perform inference

- a. For (observed or simulated) data, draw posterior samples from the trained network.
- b. Optionally importance sample to true posterior. Gives Bayesian evidence.

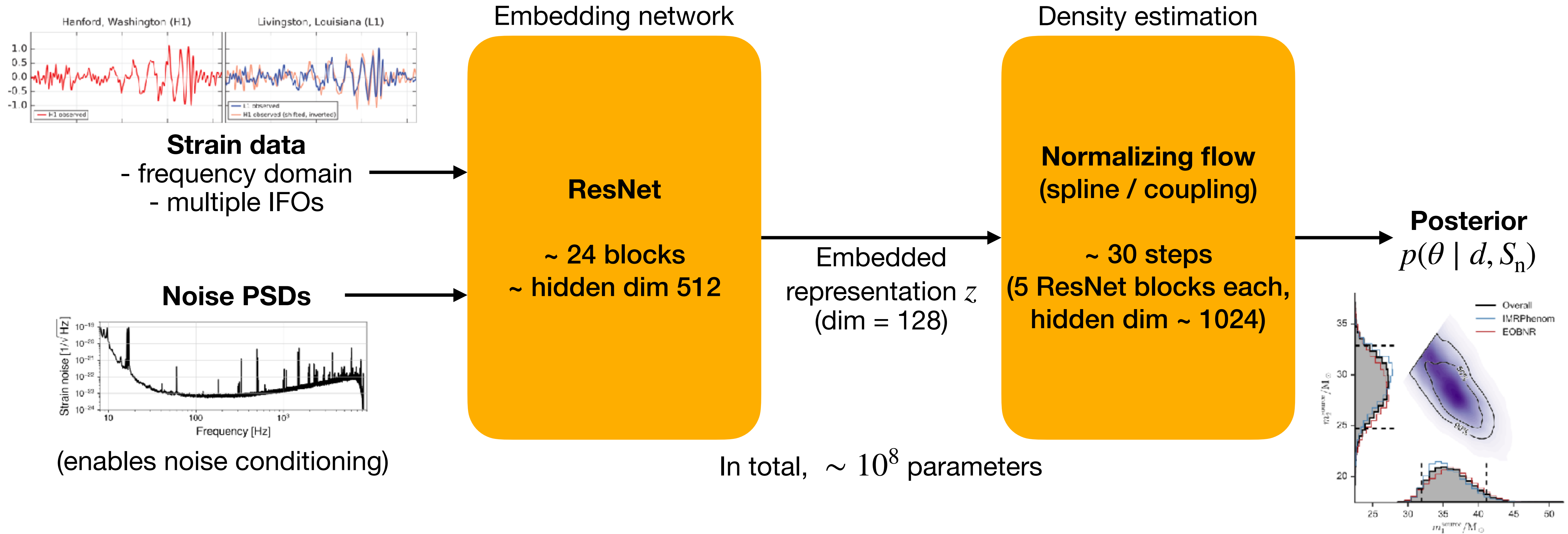
Specified by
YAML config files

Use Dingo API or
INI file

DINGO



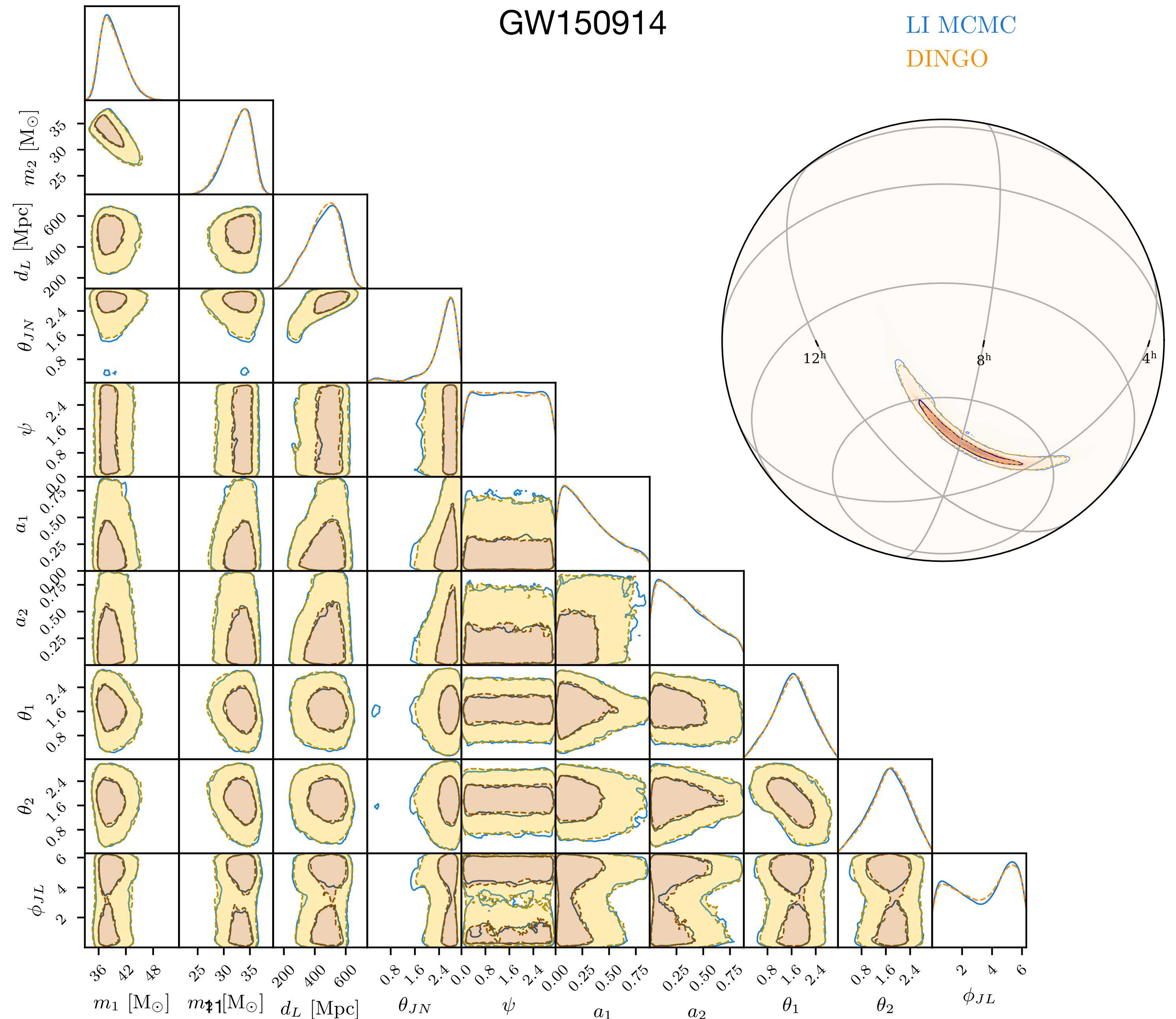
Baseline architecture



DINGO

Results

- Excellent agreement with standard samplers.
- Inference in seconds to minutes.
- Also align merger times in different detectors to simplify data representations (group equivariant NPE).



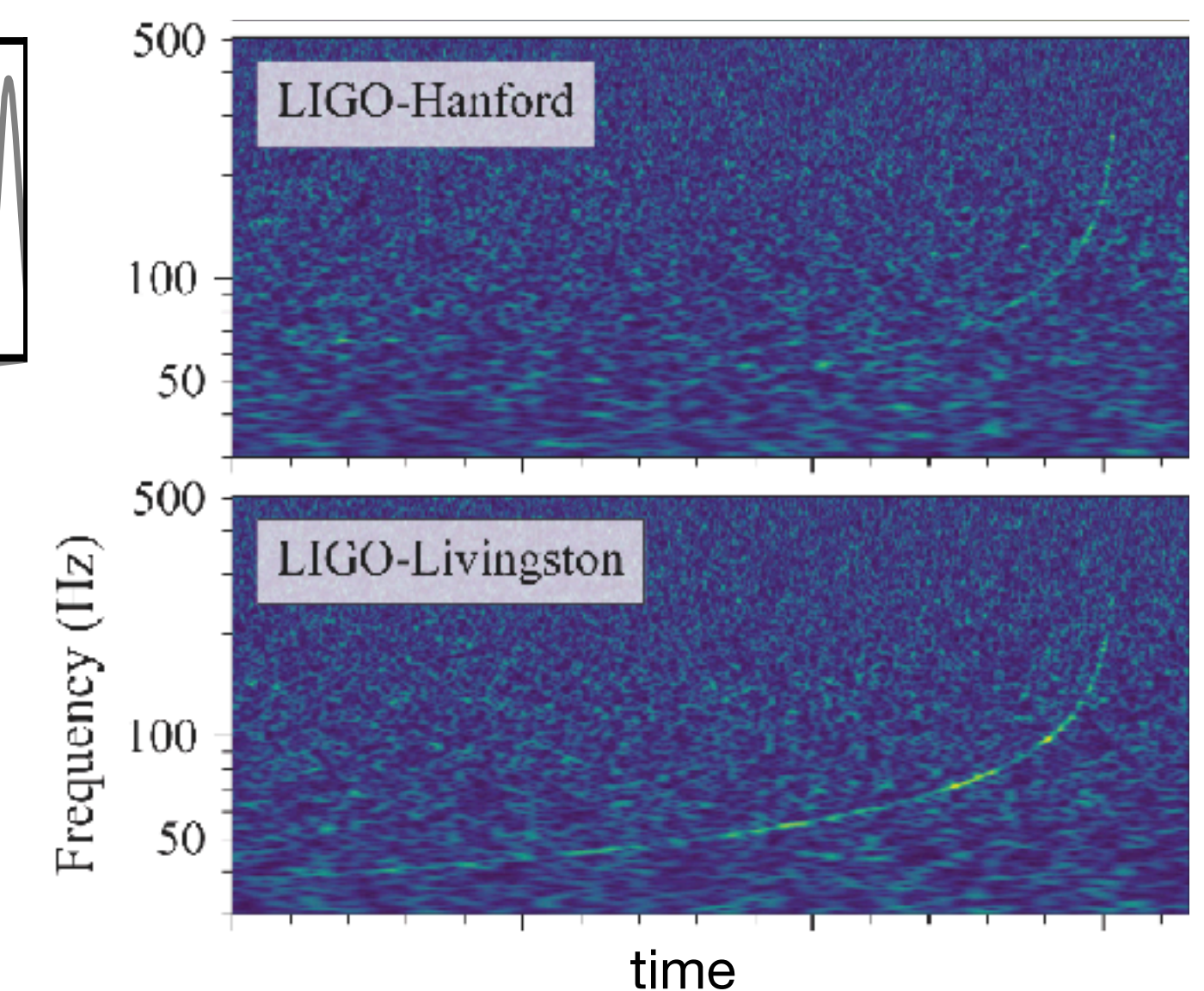
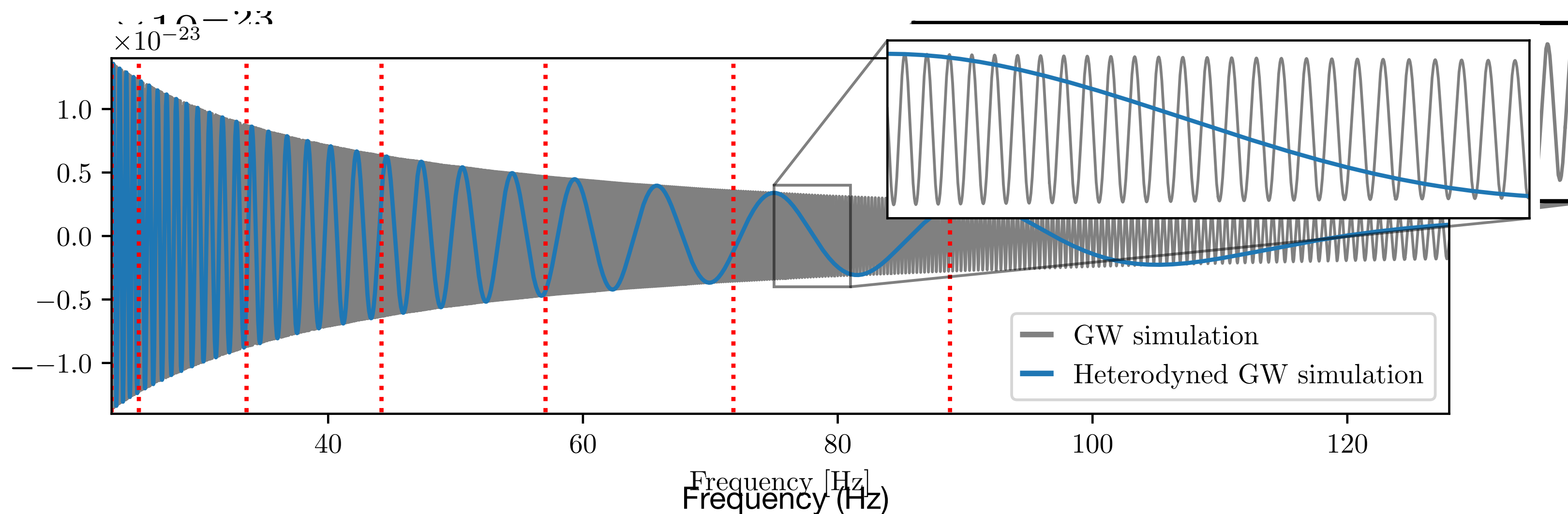
Binary neutron stars

Use physics information to simplify model

- **Heterodyning:** Factor out overall phase $\propto (\mathcal{M}f)^{-5/3}$
- **Multi-banding:** Use coarser resolution at high frequencies

Prior conditioned model

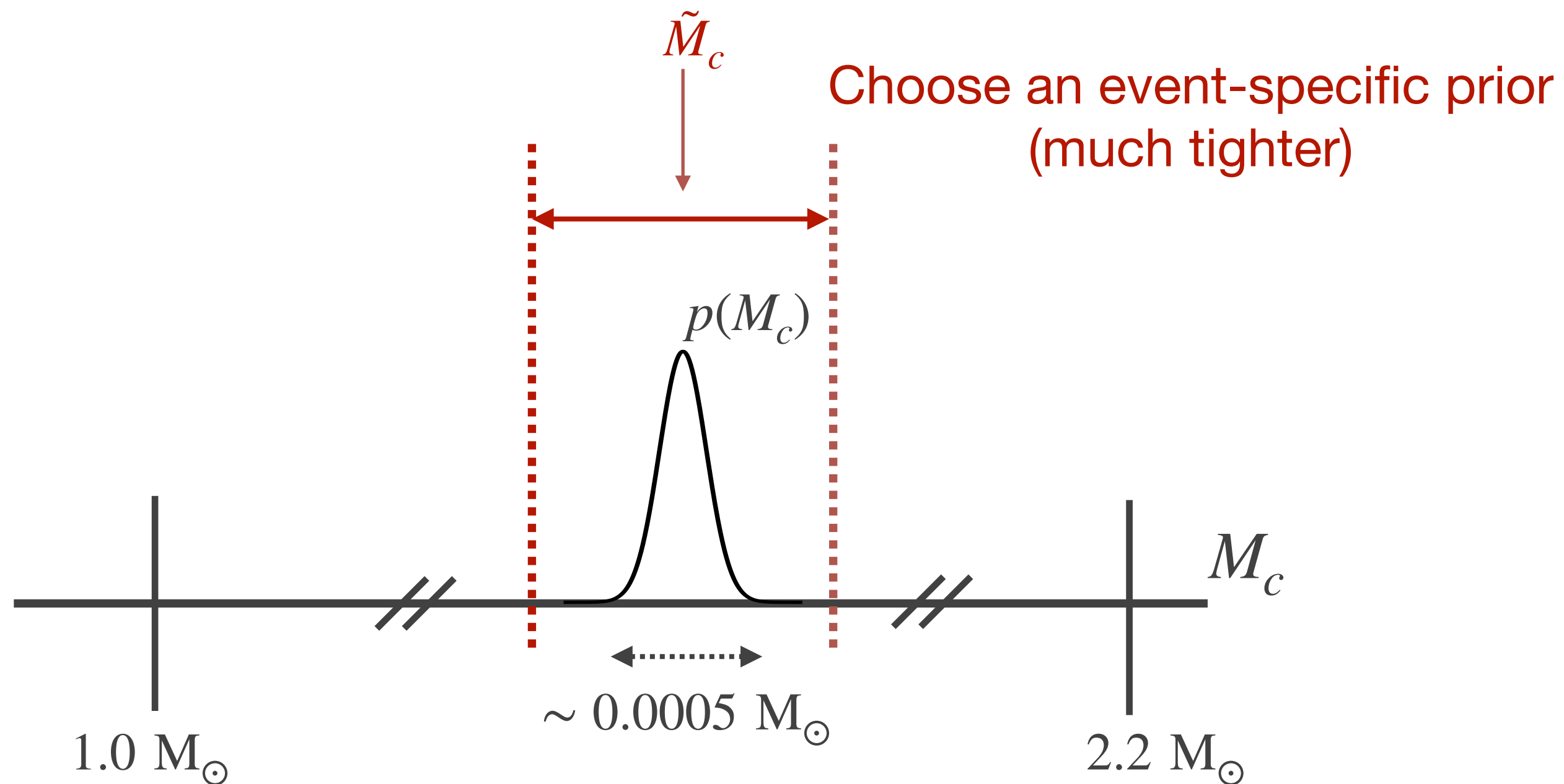
$$q(\theta \mid d \cdot e^{i\varphi_{\mathcal{M}}(f)}, \mathcal{M})$$



Binary neutron stars

Prior conditioning

- Heterodyning depends on a **chirp mass** $\tilde{M}_c$, which varies between events.



Prior conditioning

Train a network that is tunable to an **event-specific prior**

1. Select the event-specific prior $\tilde{M}_c \sim p(\tilde{M}_c)$
2. Sample (θ, d) within this prior
3. Condition the network $\rightarrow q(\theta | d, \tilde{M}_c)$
4. Heterodyne the data $d \rightarrow d_{\tilde{M}_c}$

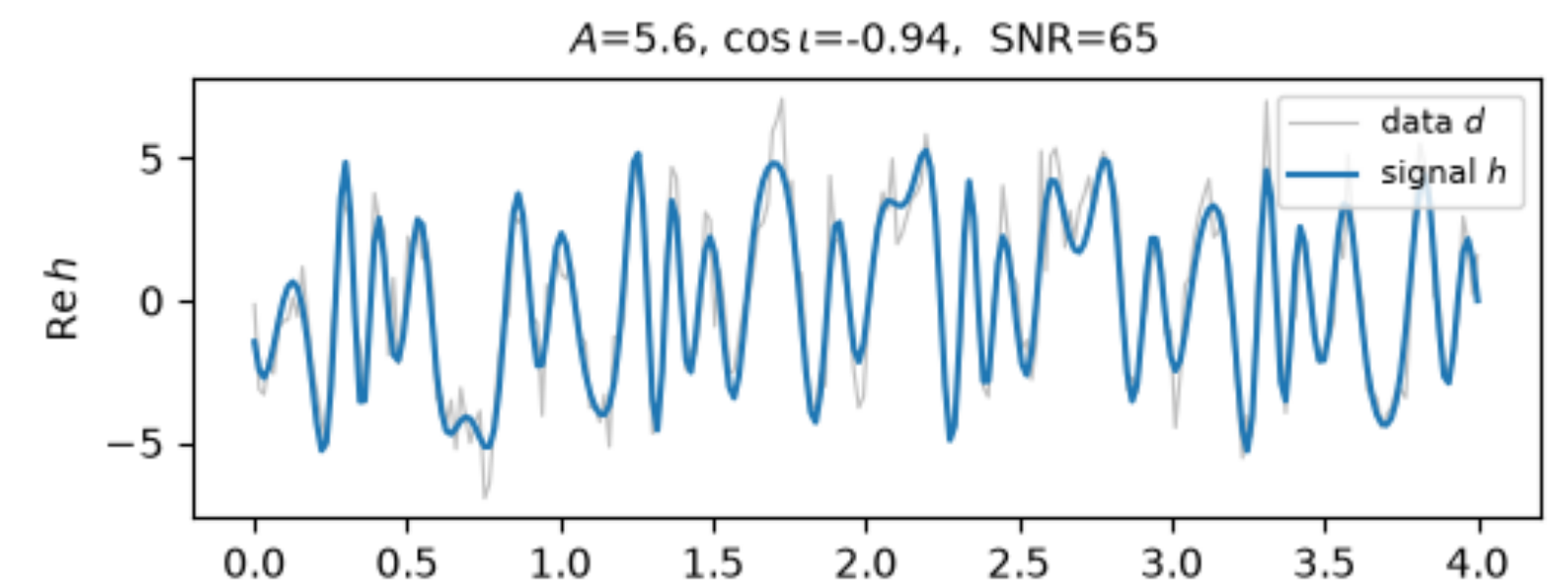
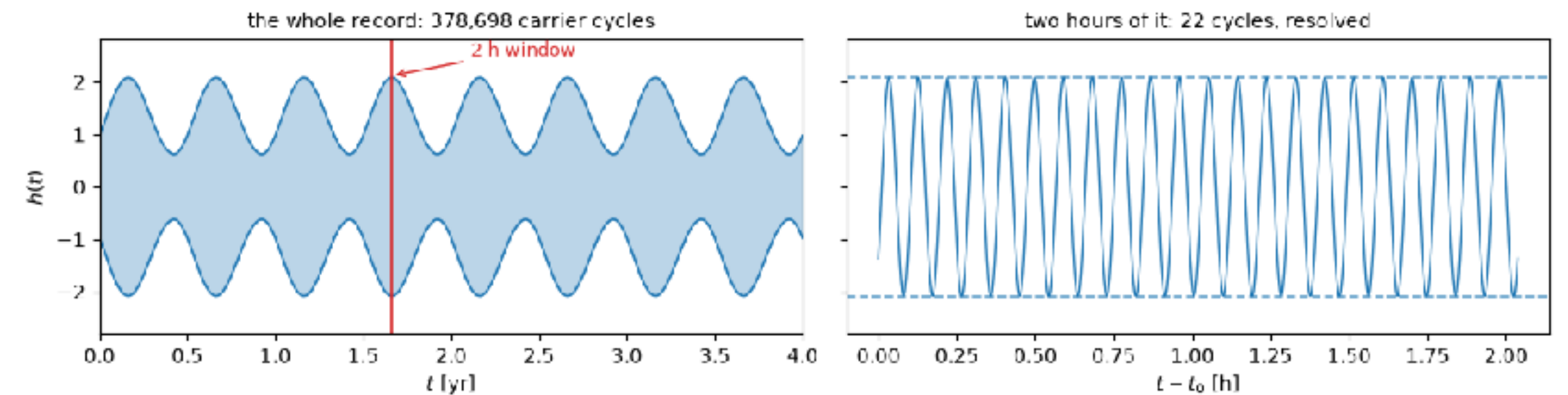
Galactic binary demo

- Consider a single galactic binary of frequency, e.g., 3 mHz. Over the LISA mission, this is hundreds of thousands of oscillations.
- Work instead with $\delta f = f - f_{\text{ref}}$ with the following toy model. Includes Doppler modulation and rotating antenna pattern

$$s_{\theta}(t) = A \left[F_{+}(t) \frac{1 + \cos^2 \iota}{2} + i F_{\times}(t) \cos \iota \right] e^{i\Phi(t)}$$

$$\Phi(t) = 2\pi \left(\delta f t + \frac{1}{2} \dot{f} t^2 \right) + \phi_0 + 2\pi f_{\text{ref}} \frac{R}{c} \cos \beta \cos \left(\frac{2\pi t}{\text{yr}} - \lambda \right)$$

$$F_{+}(t) = \frac{1}{2} \left[1 + 0.6 \cos \left(\frac{4\pi t}{\text{yr}} - 2\lambda \right) \right], \quad F_{\times}(t) = \frac{1}{2} \left[1 + 0.6 \sin \left(\frac{4\pi t}{\text{yr}} - 2\lambda \right) \right]$$



Galactic binary demo

- **Part 1:**
 - Infer parameters using normalizing flow, based on narrow δf prior (10^{-7} Hz).
 - Validate with P—P plot, MCMC, and IS
- **Part 2a:**
 - Extend prior by factor $W = 20$ and NPE fails
 - Heterodyne similarly to DINGO-BNS, and we can reuse the network from Part 1 to cover the full prior.
- github.com/stephengreen/LISAI-Hackathon-ESTEC

